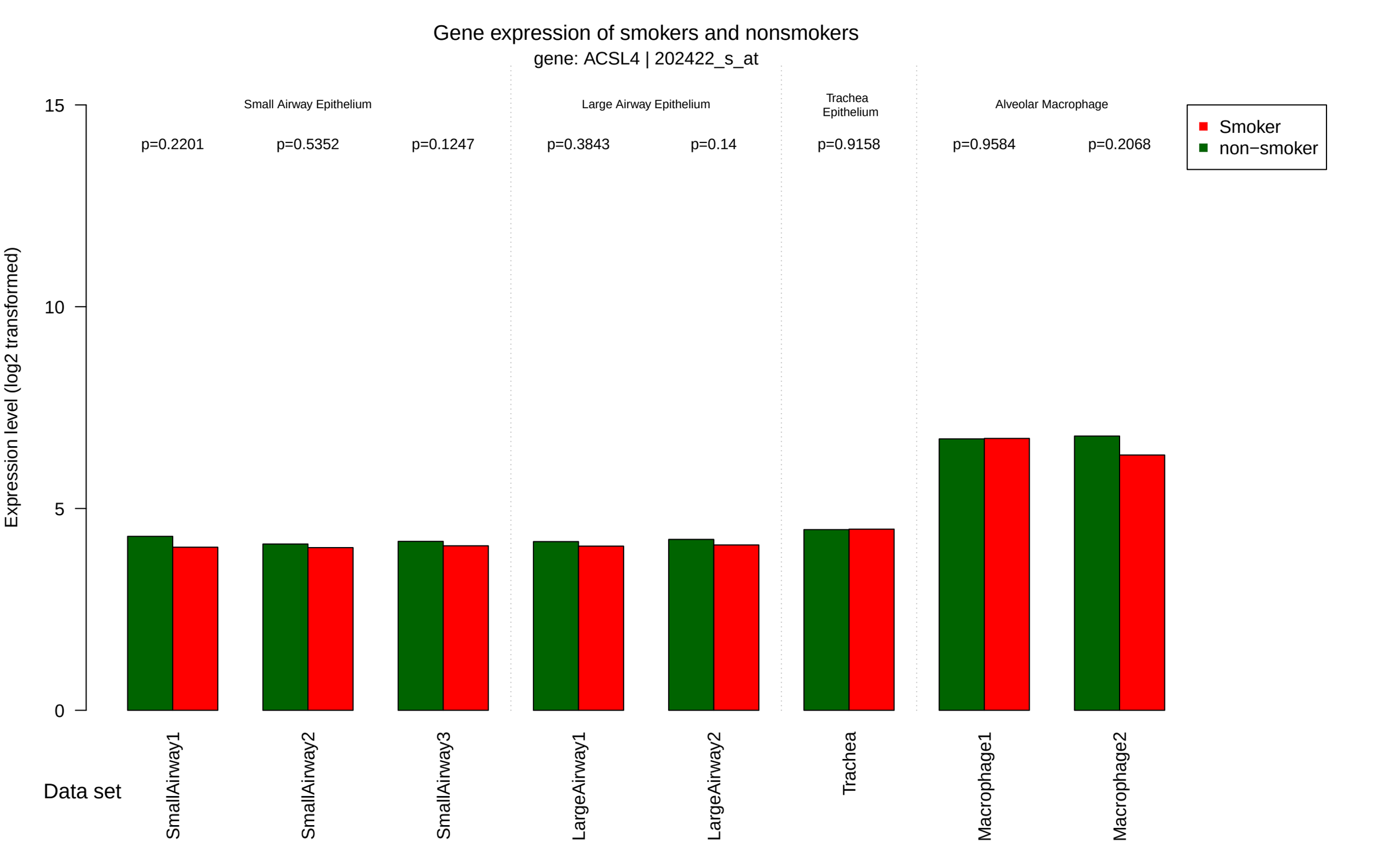
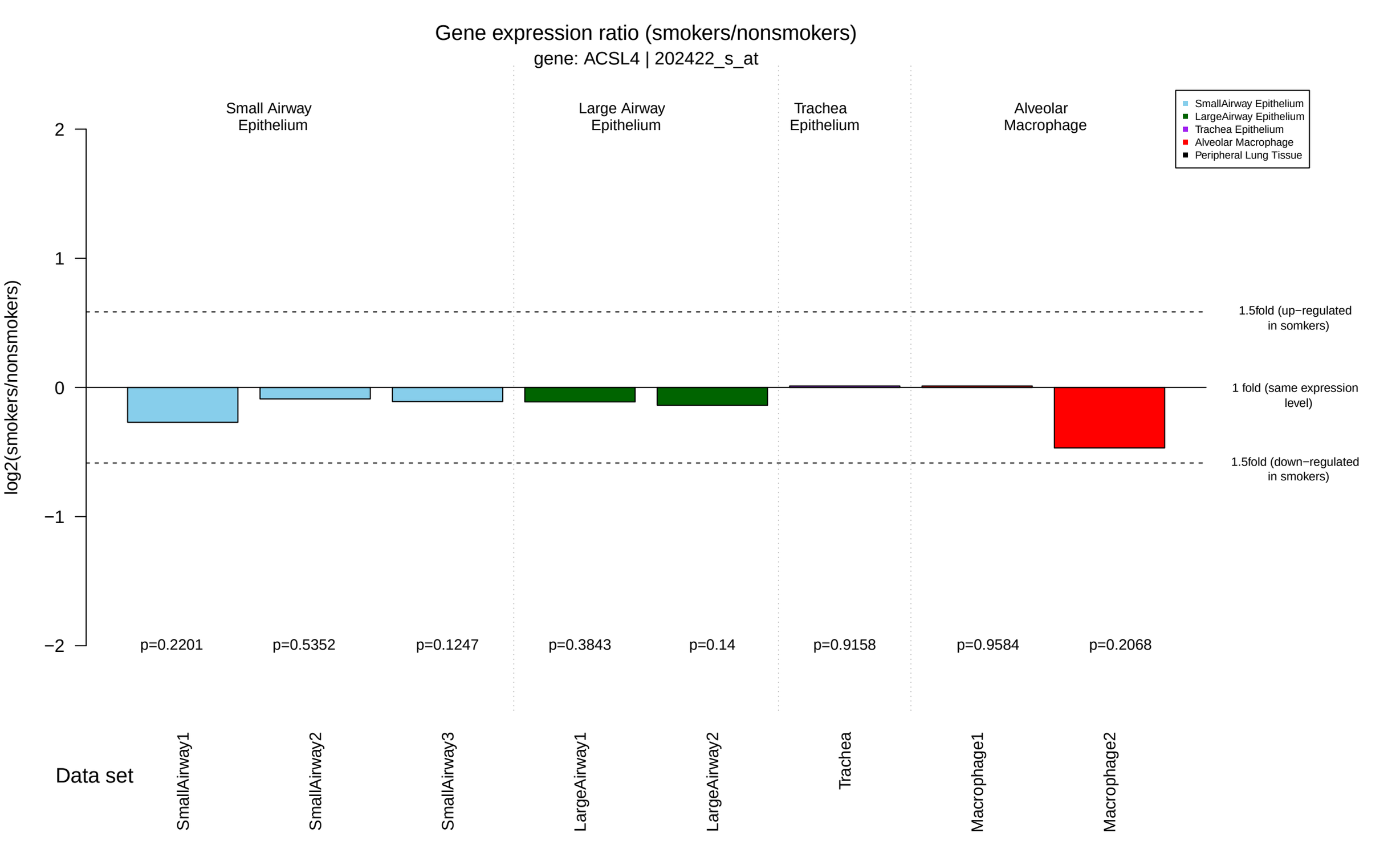
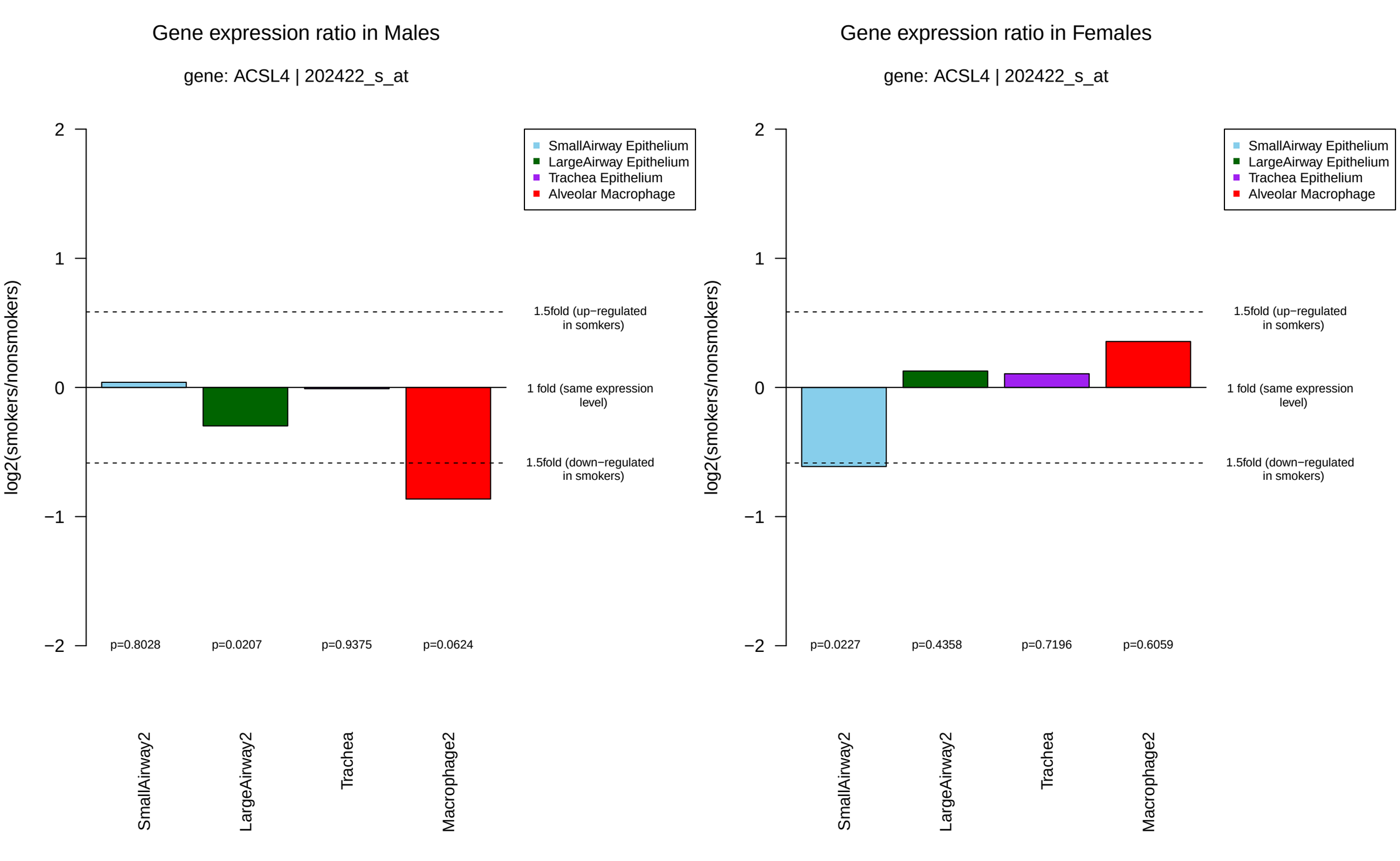
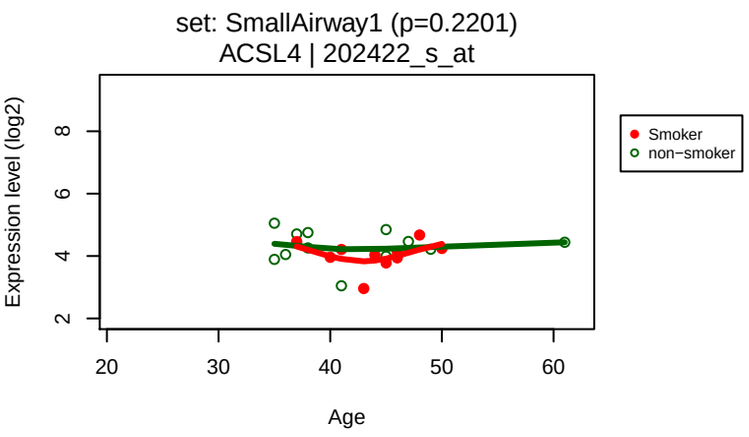
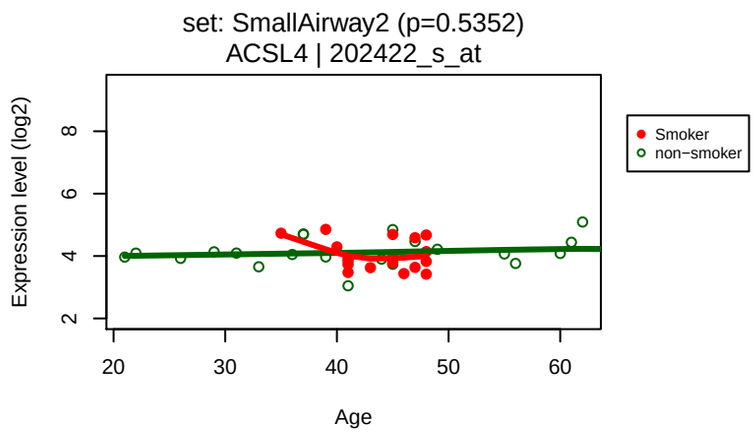
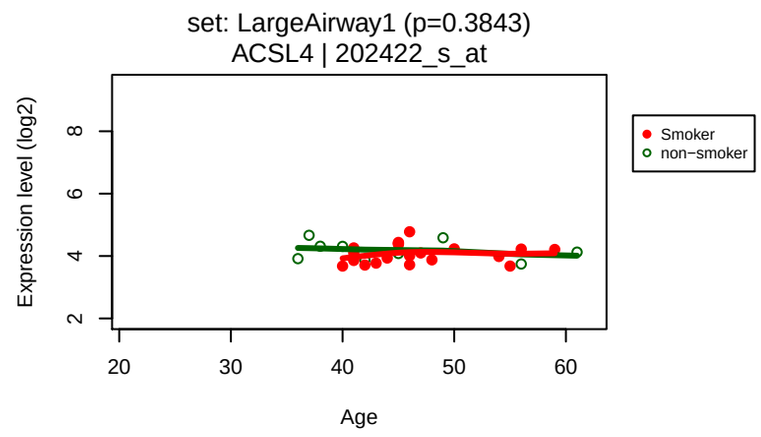

Gene expression of smokers and nonsmokers
gene: ACSL4 | 202422_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2201
p=0.5352
p=0.1247
p=0.3843
p=0.14
p=0.9158
p=0.9584
p=0.2068
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

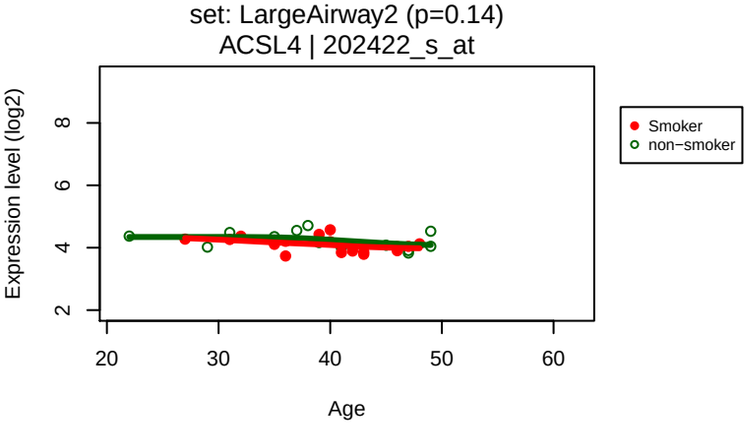

Gene expression ratio (smokers/nonsmokers)
gene: ACSL4 | 202422_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2201
p=0.5352
p=0.1247
p=0.3843
p=0.14
p=0.9158
p=0.9584
p=0.2068
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

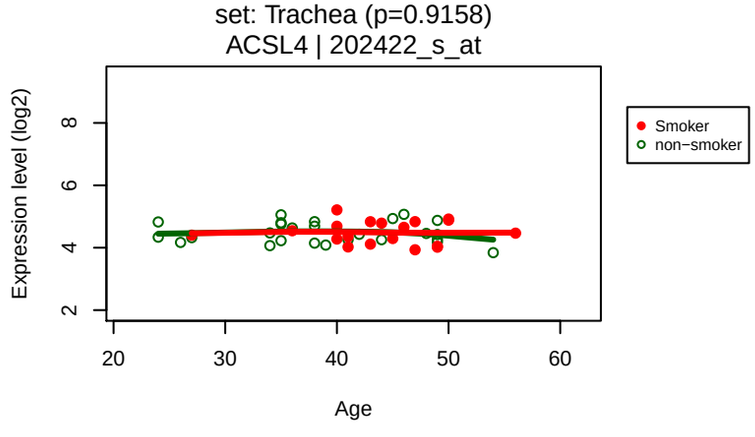

Gene expression ratio in Males
gene: ACSL4 | 202422_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8028
p=0.0207
p=0.9375
p=0.0624
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACSL4 | 202422_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0227
p=0.4358
p=0.7196
p=0.6059
SmallAirway2
LargeAirway2
Trachea
Macrophage2

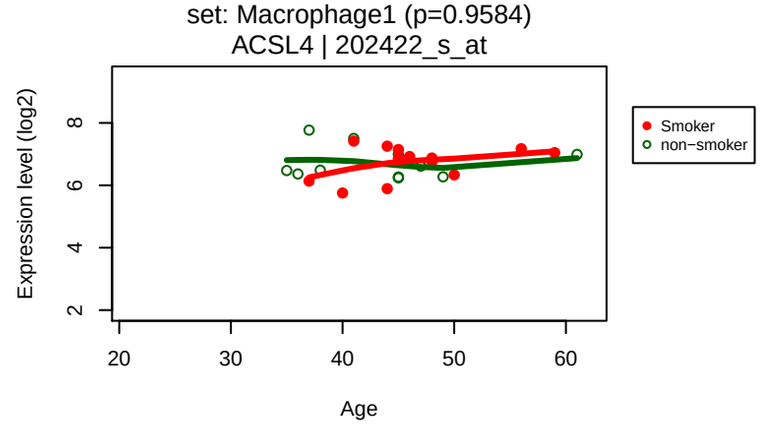

set: SmallAirway1 (p=0.2201)
ACSL4 | 202422_s_at
Expression level (log2)
Age
Smoker
non−smoker

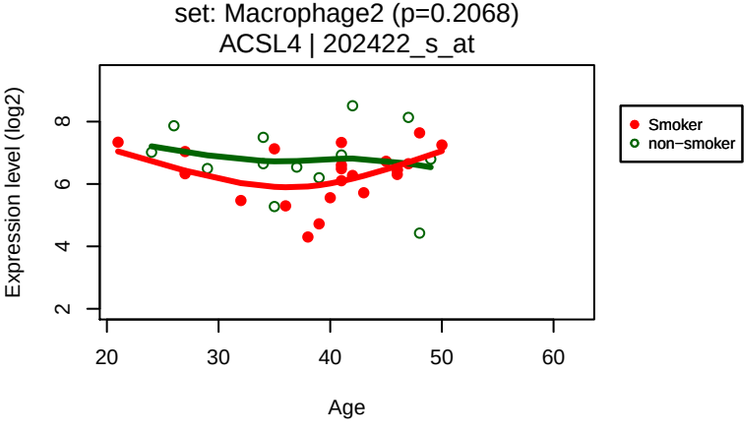

set: SmallAirway2 (p=0.5352)
ACSL4 | 202422_s_at
Expression level (log2)
Age
Smoker
non−smoker

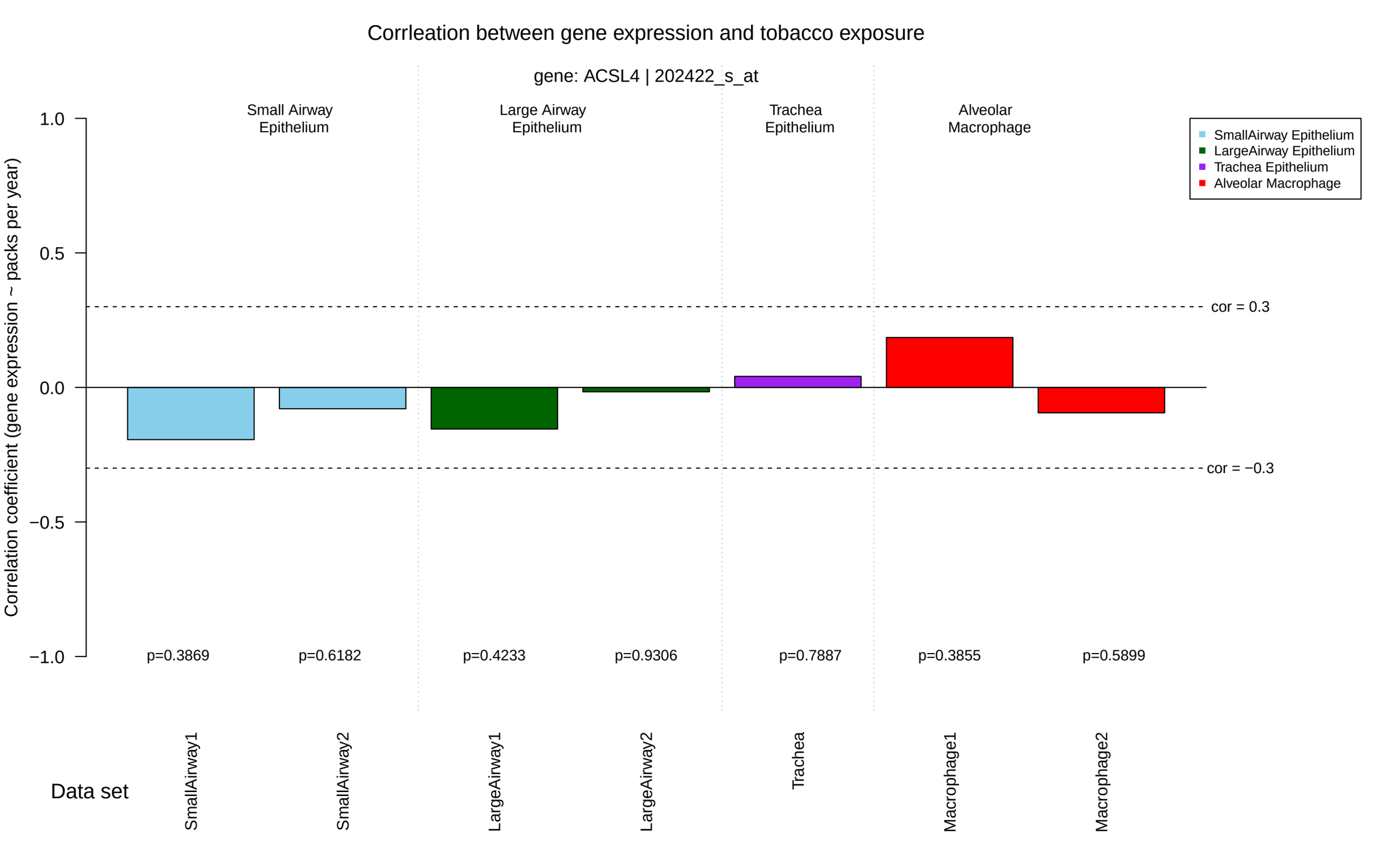

set: LargeAirway1 (p=0.3843)
ACSL4 | 202422_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.14)
ACSL4 | 202422_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9158)
ACSL4 | 202422_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9584)
ACSL4 | 202422_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2068)
ACSL4 | 202422_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACSL4 | 202422_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3869
p=0.6182
p=0.4233
p=0.9306
p=0.7887
p=0.3855
p=0.5899
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2